Data in Brief: Credit Risk Assessment Dataset with Explainable AI and Uncertainty Quantification

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Abstract

This article introduces the Home Equity (HMEQ) dataset for credit risk assessment which contains 5,960 home equity loan applications from U.S. borrowers. The data include borrower characteristics, loan terms and delinquency outcomes. The default rate is 19.9%. Data preprocessing included mean imputation for missing numerical values, mode imputation for categorical variables, one-hot encoding, and Synthetic Minority Over-sampling Technique (SMOTE) to deal with class imbalance. Three machine learning models, Logistic Regression, Random Forest and XGBoost were evaluated by predictive accuracy, expected calibration error, uncertainty quantification and explainability techniques (SHAP and LIME). Random Forest and XGBoost were able to achieve accuracy above 98% with AUC scores above 0.999, XGBoost showing better calibration (ECE = 0.0117). The data are publicly available and allow for research on transparent uncertainty-aware credit scoring systems.

Specifications Table

Table 1: Specifications Table.

Subject area	Financial Risk Modelling, Machine Learning
More specific subject area	Credit Risk Assessment, Explainable AI (XAI), Uncertainty Quantification (UQ)
Type of data	CSV file (tabular data)
How data were acquired	Publicly downloaded from ListenData repository (https://www.listendata.com/2019/08/datasets-for-credit-risk-modeling.html)
Data format	Raw and preprocessed
Experimental factors	Class imbalance (19.9% default rate); missing values in numerical and categorical variables
Experimental features	Preprocessing: mean/mode imputation, one-hot encoding, SMOTE balancing; Modelling: Logistic Regression, Random Forest, XGBoost; Evaluation: accuracy, precision, recall, F1-score, ROC-AUC, ECE, entropy-based uncertainty
Data source location	United States (home equity loan applicants)
Data accessibility	Publicly available at https://www.listendata.com/2019/08/datasets-for-credit-risk-modeling.html?m=1
Related research article	Rambauli, M., Ravele, T., & Sigauke, C. (2026). Application of Explainable AI and Uncertainty Quantification in Credit Risk Assessment. *Journal* (submitted).

Value of the Data

- The dataset provides a standardised benchmark (HMEQ) for comparing traditional statistical methods (Logistic Regression) with ensemble machine learning approaches (Random Forest, XGBoost) in default prediction.
- With 19.9% default rate, the dataset is suitable for evaluating resampling techniques such as SMOTE and their impact on minority class detection.
- The dataset enables testing of SHAP and LIME methods for both global and local interpretability in regulated financial domains requiring decision transparency.
- Rich probability outputs support evaluation of calibration metrics (ECE) and entropy-based uncertainty quantification, critical for probability of default estimation and risk-based pricing.
- The dataset facilitates research on GDPR Article 22 compliance ("right to explanation") and model governance in automated lending systems.

1 Data

1.1 Dataset Description

The Home Equity (HMEQ) dataset contains **5,960 observations** of home equity loan applicants from the United States. Each record represents a loan application with **20 variables** capturing borrower demographics, financial history, loan characteristics, and default outcomes.

1.2 Target Variable

The binary target variable `BAD` indicates loan default status:

- **1** = Borrower defaulted or became seriously delinquent
- **0** = Borrower repaid the loan as agreed

Class distribution:

- Default (BAD = 1): 1,189 cases (19.9%)
- Non-default (BAD = 0): 4,771 cases (80.1%)

This represents a moderate class imbalance (approximately 1:4 ratio).

1.3 Predictor Variables

Table 2: Predictor variables in the HMEQ dataset.

Variable	**Description**	**Type**
LOAN	Loan amount requested	Numerical
MORTDUE	Balance on existing mortgage	Numerical
VALUE	Current property value	Numerical
REASON	Loan purpose (DebtCon = debt consolidation; HomeImp = home improvement)	Categorical
JOB	Occupation type (Mgr, Office, Other, ProfExe, Sales, Self)	Categorical
YOJ	Years in current job	Numerical
DEROG	Number of major derogatory credit reports	Numerical
DELINQ	Number of delinquent credit lines	Numerical
CLAGE	Age of oldest credit line (months)	Numerical
NINQ	Number of recent credit inquiries	Numerical
CLNO	Total number of credit lines	Numerical
DEBTINC	Debt-to-income ratio (%)	Numerical

1.4 Sample Records

Table 3: Sample records from the HMEQ dataset.

BAD	LOAN	MORTDUE	VALUE	REASON	JOB	YOJ	DEROG	DELINQ	CLAGE	NINQ	CLNO	DEBTINC
1	1700	97801	112000	HomeImp	Office	30	0	0	93.33	0	14	NaN
1	1700	305484	320000	HomeImp	Other	9	0	0	101.47	1	8	37.11
1	1800	48649	57037	HomeImp	Other	5	3	2	77.10	1	17	NaN
0	3500	132800	175000	DebtCon	Mgr	11	0	0	290.87	1	23	29.41
0	2300	140200	195600	DebtCon	ProfExe	10	0	0	243.63	2	35	31.92

1.5 Performance Metrics

Table 4: Performance metrics.

Metric	**Formula**
Accuracy	(TP + TN) / (TP + TN + FP + FN)
Precision	TP / (TP + FP)
Recall	TP / (TP + FN)
F1-Score	2 × (Precision × Recall) / (Precision + Recall)
ROC-AUC	Area under TPR vs. FPR curve
ECE	$\sum_{k=1}^{K} \frac{\lvert S_k \rvert}{T} \lvert \hat{y}_k - \hat{p}_k \rvert$

1.6 Software and Packages

Table 5: Software and packages used.

Purpose	Software/Package
Data manipulation	Pandas, NumPy
Visualisation	Matplotlib, Seaborn
Machine learning	Scikit-learn (LR, RF), XGBoost
Explainable AI	SHAP, LIME
Imbalanced learning	Imbalanced-learn (SMOTE)
Programming language	Python 3.x

2 Results Summary

Table 6: Model performance comparison.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	ECE
Random Forest	0.9866	0.9906	0.9825	0.9865	0.9992	0.0475
XGBoost	0.9854	0.9941	0.9767	0.9853	0.9991	0.0117
Logistic Regression	0.7739	0.7765	0.7692	0.7728	0.8649	0.0330

2.1 Key Findings

1. **Ensemble methods** (RF, XGBoost) significantly outperform logistic regression, achieving $>$98% accuracy and near-perfect AUC ($>$0.999).
2. **XGBoost provides best calibration** (ECE = 0.0117) vs. Random Forest's overconfidence (ECE = 0.0475).
3. **Top risk predictors** identified by SHAP and LIME: DELINQ, DEROG, DEBTINC.
4. **Predictive entropy** reliably distinguishes correct predictions (low entropy) from misclassifications (high entropy, 0.4–0.7 range for LR).
5. **Defaulters exhibit higher risk indicators**: 1.23 vs. 0.25 delinquencies, 0.71 vs. 0.13 derogatory records, 39.39% vs. 33.25% debt-to-income ratio.

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[1] HMEQ Dataset. ListenData. Available at: https://www.listendata.com/2019/08/datasets-for-credit-risk-modeling.html (Accessed: 13 February 2025).